1. 积分 (0在最後面)

电生磁: $\oint \vec{H}\, d\vec{l} = \iint (\vec{J}+\frac{\partial \vec{D}}{\partial t})\, d\vec{s}$

磁生电: $\oint \vec{E}\, d\vec{l} = -\frac{\partial}{\partial t}\iint \vec{B}\, d\vec{s}$

有电荷: $\oiint \vec{D}\, d\vec{s} = \iiint \rho\, dv$

没磁荷: $\oiint \vec{B}\, d\vec{s} = 0$

2. 微分

电生磁: $\nabla\times\vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$

磁生电: $\nabla\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}$

有电荷: $\nabla\cdot\vec{D} = \rho$

没磁荷: $\nabla\cdot\vec{B} = 0$

考完试，一定要回頭看，是不是每个矢量，都打了箭头！

3. 要推 $\nabla\cdot\vec{J} = -\frac{\partial \rho}{\partial t}$

$$0 = \nabla\cdot(\nabla\times\vec{H})$$
$$= \nabla\cdot\vec{J} + \nabla\cdot\left(\frac{\partial \vec{D}}{\partial t}\right)$$
$$= \nabla\cdot\vec{J} + \frac{\partial(\nabla\cdot\vec{D})}{\partial t} = \nabla\cdot\vec{J} + \frac{\partial \rho}{\partial t}$$

仅供参考，若有不同之處
请相信自己

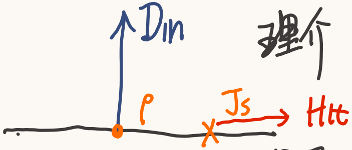
21 YYY

4 $\oiint (\vec{E} \times \vec{H}) \cdot d\vec{S} + \iiint \vec{J} \cdot \vec{E}\, dV = -\frac{\partial}{\partial t} \iiint \left(\frac{1}{2} \vec{B} \cdot \vec{H} + \frac{1}{2} \vec{E} \cdot \vec{D} \right) dV$

流出功率流　　电能损耗　　电磁能量减少

具体定义翻书

5



理介

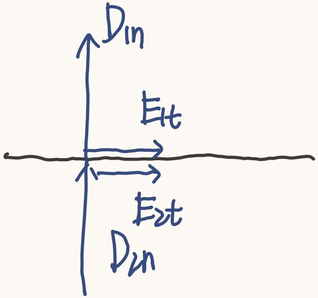
理导

$$\begin{cases} D_{1n} = \rho_s \\ E_{1t} = E_{2t} = 0 \end{cases}$$

注意矢量，或写成 $\vec{n} \cdot \vec{D_1} = \rho_s$

$$\begin{cases} H_{1t} = J_s \\ B_{1n} = B_{2n} = 0 \end{cases}$$

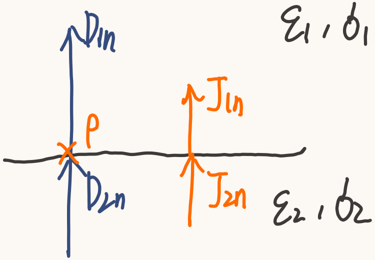
6



$$\begin{cases} D_{1n} = D_{2n} \\ E_{1t} = E_{2t} \end{cases}$$

D_n 连续不是 E_n 连续

$$\begin{cases} H_{1t} = H_{2t} \\ B_{1n} = B_{2n} \end{cases}$$

7 

$$\begin{cases} D_{1n}-D_{2n}=\rho \\ J_{1n}=J_{2n} \end{cases} \Rightarrow \begin{cases} \varepsilon_1 E_{1n}-\varepsilon_2 E_{2n}=\rho \\ \sigma_1 E_{1n}=\sigma_2 E_{2n} \end{cases}$$

8 所有 E_n 写成 $-\frac{\partial\varphi}{\partial n}$ $(-\nabla\varphi)$

所有 B 写成 $\nabla\times\vec{A}$

$\varphi_1=\varphi_2 \Leftrightarrow E_{t1}=E_{t2}$

$A_{1n}=A_{2n} \Leftrightarrow \nabla\cdot\vec{A}=0$ (库伦规范)

9 $D=\varepsilon E=\varepsilon_0\varepsilon_r E=\varepsilon_0 E+\varepsilon_0\underbrace{(\varepsilon_r-1)}_{\chi_e}E=\varepsilon_0 E+P$

$\sigma_{ps}=\vec{P}\cdot\vec{n}$

$\sigma_{内}=-\nabla\cdot\vec{P}$

10 $B=\mu H=\mu_0\mu_r H=\mu_0 H+\mu_0\underbrace{(\mu_r-1)}_{x_m}H=\mu_0 H+\mu_0 M$

$\vec{J}_{ms}=\vec{M}\times\vec{n}$

$\vec{J}_{体}=\nabla\times\vec{M}$

$\nabla\times H=J+\frac{\partial D}{\partial t}$

11 $\nabla\varphi=-\vec{E}$

$\nabla\cdot\nabla\varphi=-\nabla\cdot\vec{E}$

$\nabla^2\varphi=\begin{cases}0\\-\frac{\rho}{\varepsilon}\end{cases}$

$\nabla\times\vec{A}=\vec{B}$

$\nabla\times\nabla\times\vec{A}=\nabla(\nabla\cdot A)-\nabla^2A$

$\nabla\times\vec{B}=\nabla 0-\nabla^2A$

$\nabla^2A=\begin{cases}0\\-\mu J\end{cases}$

12 一个矢量场可以由其散度和旋度唯一确定
一个矢量场可分解成一个无旋场和一个无散场
(结合边界条件)

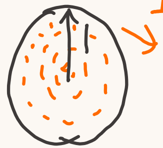
13 标量场梯度无旋

矢量场旋度无散

14 $\vec{p} = q\cdot\vec{l}$

15 对. 理导电流只在表面

→ $I=1A$. $J=\frac{1}{2\pi}A$



$I=1A$

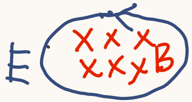
无限薄之处，电流大小为0 $J=\frac{0}{2\pi}=0$

16 $\varphi_1=\varphi_2 \leftrightarrow E_{1t}=E_{2t}$

17 对。无"磁荷"

18 错。静电场不闭合。 $\oint \vec{E}\cdot d\vec{l} = -\frac{\partial}{\partial t}\int B\cdot d\vec{S}$

时变可能闭合。



发电机

19 ✓

20 有

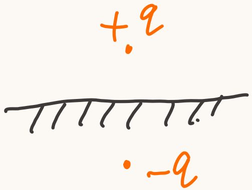
21 ε和结构，与E,H无关　　μ与结构，与E,H无关

22 $w_e = \frac{1}{2}\vec{D}\cdot\vec{E}$　　$w_m = \frac{1}{2}\vec{B}\cdot\vec{H}$ （密度），可以

23

恒流	静
E恒	E静
φ	φ
J	D
I	Q
σ	ε
G	C

24

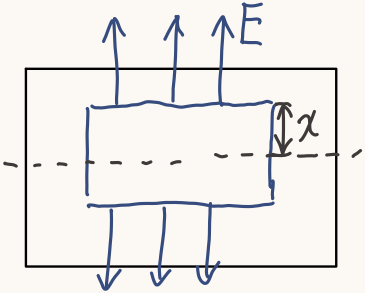


25 查书

XXX，知道所有 $\left\{ \begin{array}{l} \varphi \\ \frac{\partial \varphi}{\partial n} \end{array} \right.$，xxx 唯一

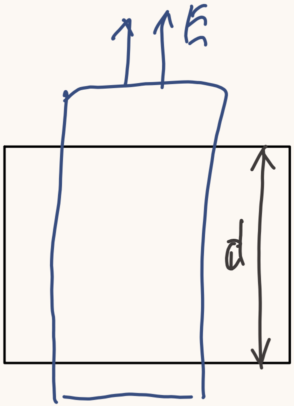
26 $\vec{J} = \vec{J}_{ct} \ \vec{J}_{v} + \frac{\partial \vec{D}}{\partial t}$

小题



$2DL = 2x\rho l$

$E = \frac{\rho}{\varepsilon} \cdot x$



$2DL = d\rho l$

$E = \frac{\rho}{2\varepsilon} \cdot d$

2. $D\cdot 4\pi r^2 = q$

$\vec{E} = \frac{q}{4\pi\varepsilon r^2}\cdot\vec{a_r}$ 写的时候带矢量

3 $D\cdot 2\pi r = \rho$

$E = \frac{\rho}{2\pi\varepsilon r}$

4 $2DS = \rho S$

$E = \frac{\rho}{2\varepsilon}$

5 $H\cdot 2\pi r = I$

$B = \frac{\mu I}{2\pi r}$

大题一

(1) $\vec{E} = \frac{q}{4\pi\varepsilon r}\cdot\vec{a_r}$

$U = \int_a^b \frac{q}{4\pi\varepsilon r}dr = \frac{q}{4\pi\varepsilon}\ln\frac{b}{a}$

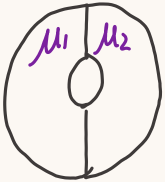
$C = \frac{q}{U}$

(2)

设流过I， $J=\frac{I}{\pi a^2}$

$$H\cdot 2\pi r = J\cdot \pi r^2$$

$$\Rightarrow B=\frac{\mu I}{2\pi a^2}\cdot r$$



的情况

$$H\cdot 2\pi r = I$$

$B_1=\mu_1 H_1$ ， $B_2=\mu_2 H_2$

(3)

$$H\cdot 2\pi r = I$$

$$B=\frac{\mu I}{2\pi r}$$

$$\Phi=\int_a^b B\cdot dl = \frac{\mu I}{2\pi}\ln\frac{b}{a}$$

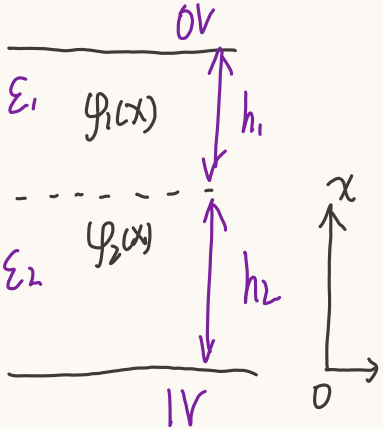
$$L=\frac{\Phi}{I}$$

(4)

$$J=\frac{-I}{(\pi c^2-\pi b^2)}$$

$$H\cdot 2\pi r = J\cdot(\pi r^2-\pi b^2)+I$$

大题二

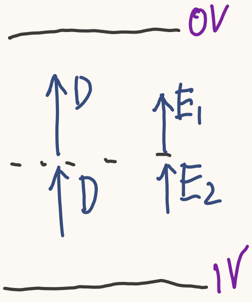


电位方程 $\begin{cases} \nabla^2\varphi_1 = 0 \\ \nabla^2\varphi_2 = 0 \end{cases} \Rightarrow \begin{array}{l} \varphi_1 = Ax+B \\ \varphi_2 = Cx+D \end{array}$

边界条件 $\begin{cases} \varphi_1(h_2) = \varphi_2(h_2) \\ \varepsilon_1\left(-\frac{\partial\varphi_1}{\partial n}\right) = \varepsilon_2\left(-\frac{\partial\varphi_2}{\partial n}\right)\Big|_{x=h_2} \end{cases}$

$\begin{cases} \varphi_2(0) = 1V \\ \varphi_1(h_1+h_2) = 0V \end{cases}$

$E = -\frac{\partial\varphi}{\partial n}$



别解电位方程、

$\begin{cases} \varepsilon_1 E_1 = \varepsilon_2 E_2 \\ E_1 h_1 + E_2 h_2 = 1V \end{cases}$

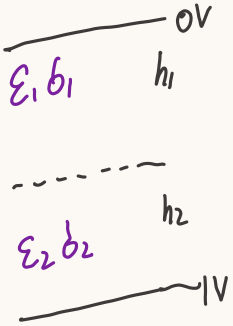
解这个

面电荷用 $D_n = \rho$

注意方向.

外面无E

大题二 困难版



$$\begin{cases}\nabla^2\varphi_1=0\\ \nabla^2\varphi_2=0\end{cases} \Rightarrow \sim$$

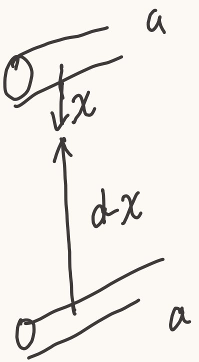
$$\begin{cases}\varphi_2(0)=1, \quad \varphi_1=\varphi_2\Big|_{x=\frac{d}{2}}\\ \varphi_1(h_1+h_2)=0\end{cases}$$

$$\begin{cases}\varepsilon_1\left(-\dfrac{\partial\varphi_1}{\partial n}\right)-\varepsilon_2\left(-\dfrac{\partial\varphi_2}{\partial n}\right)=\rho_{中间}\Big|_{x=\frac{d}{2}}\\ \sigma_1\left(-\dfrac{\partial\varphi_1}{\partial n}\right)=\sigma_2\left(-\dfrac{\partial\varphi_2}{\partial n}\right)\Big|_{x=\frac{d}{2}}\end{cases}$$

解
$$\begin{cases}\sigma_1E_1=\sigma_2E_2\\ h_1E_1+h_2E_2=1V\end{cases}$$

$$\rho_{中间}=\varepsilon_1E_1-\varepsilon_2E_2$$

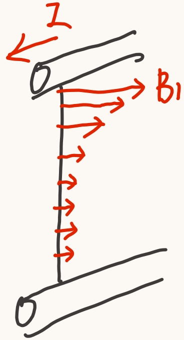
大题三

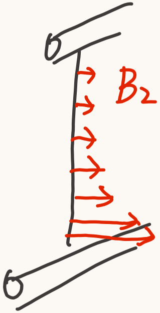


$$H_1 \cdot 2\pi x = I$$

$$H_2 \cdot 2\pi (d-x) = I$$

$$\Rightarrow B_{总} = B_1 + B_2$$



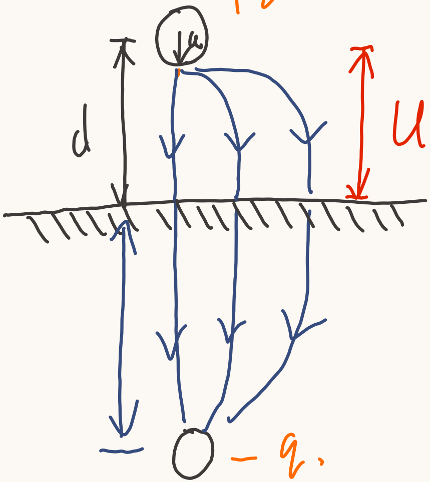


$$\Phi_m = 2\int_a^{d-a} \frac{\mu I}{2\pi x} dx$$

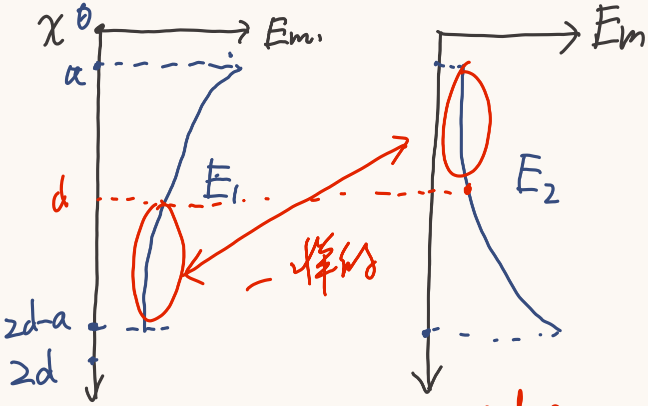
$$\approx \frac{\mu I}{2\pi} \ln \frac{b}{a}$$

$$L = \frac{\Phi_m}{I}$$

大题四



$$E = \frac{q}{2\pi \varepsilon r}$$



$$U = \int_a^d (E_1 + E_2)\,dx = \int_a^{2d-a} E_1\,dx$$

$$C = \frac{Q}{U}$$

大题五.

1. $\vec{J_d} = \frac{\partial D}{\partial t} = \varepsilon \frac{\partial E}{\partial t} = \vec{a_y} \cdot \varepsilon E_0 \sin(\frac{\pi}{d} z)(-w(\sin wt - k_x \cdot x))$

2. $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \rightarrow -\int \nabla \times \vec{E}\, dt = \int \frac{\partial \vec{B}}{\partial t} dt = \vec{B}$

$$\nabla \times \vec{E} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_y & 0 \end{vmatrix}$$

$$-\nabla \times \vec{E} = \vec{a_x} \cdot \frac{\partial E_y}{\partial z} - \vec{a_z} \cdot \frac{\partial E_y}{\partial z}$$

$$= \vec{a_x} \cdot \frac{\pi}{d} E_0 \cos\left(\frac{\pi}{d} z\right) \cos(wt - k_x \cdot x)$$

$$- \vec{a_z} \; k_x \; E_0 \sin\left(\frac{\pi}{d} z\right) \sin(wt - k_x x)$$

$$\vec{B} = \vec{a_x} \cdot \frac{\pi}{dw} E_0 \cos\left(\frac{\pi}{d} z\right) \sin(wt - k_x x)$$

$$- (-1) \vec{a_z} \; \frac{k_x}{w} E_0 \sin\left(\frac{\pi}{d} z\right) \cos(wt - k_x x)$$

注：写完一定检查，是否每个矢量打了箭头

Del 算子：$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$ 是個"矢量"

$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$ 梯度 grad

$\nabla \cdot \vec{A} = \frac{\partial A}{\partial x} + \frac{\partial A}{\partial y} + \frac{\partial A}{\partial z}$ 散度 division

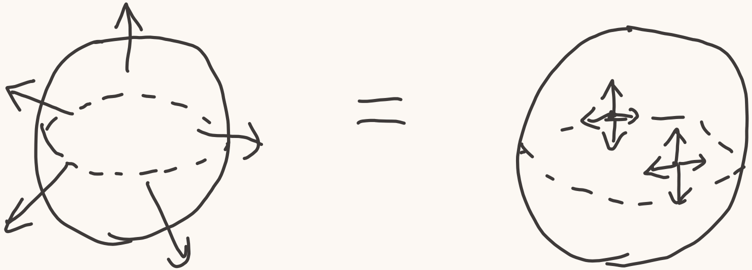
$\nabla \times \vec{A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$ 旋度 rotation

梯度是势函数变化最快的方向

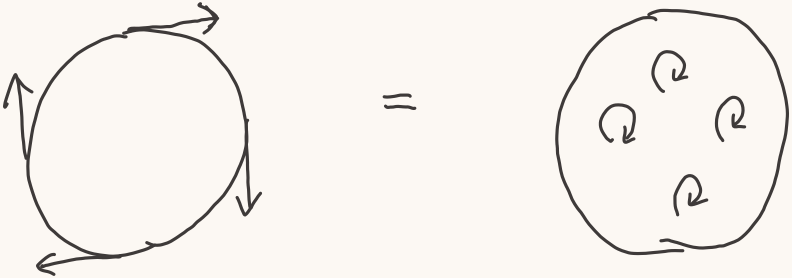
散度代表发散源，灯泡，"凭空产生了新矢量"

旋度代表旋涡源，电风扇，"绕一圈凭空可以做功了"

高斯：



斯托克斯



Laplace：$\nabla^2 = \left(\frac{\partial^2}{\partial x^2}, \frac{\partial^2}{\partial y^2}, \frac{\partial^2}{\partial z^2}\right)$